

Doob's Proof of MLE Strong Consistency: A Surprisal and Information Perspective

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Abstract

We give a concise exposition of Doob's proof of strong maximum likelihood consistency, formulated in terms of surprisal and relative information. This formulation clarifies the roles of identifiability, properness, convergence, and continuity, and highlights that consistency holds without uniform convergence of the log-likelihood.

Keywords: maximum likelihood estimator, true parameter, relative information, autoregressive, markov process, law of large numbers

1 Introduction

The consistency of the [Fisher \(1922\)](#) maximum likelihood estimator (MLE) is a classical result discovered more than a century ago. This method is typically treated as a special case of M -estimators, and derived via techniques dating back to [Cramér \(1946\)](#) and [Wald \(1949\)](#). A modern treatment is [van der Vaart \(1998\)](#). Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found the [Doob \(1934\)](#) strong consistency proof of MLE convergence more motivated and accessible to students, particularly for students already familiar with information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter spaces and general state spaces with essentially no extra effort only adds to its appeal.

The contribution of this note is expository and pedagogical: We give a formulation of Doob's consistency argument in surprisal/information terminology. Following Doob, this is done without assuming uniform convergence.

2 Surprisal and Information

A random variable x is sampled repeatedly, resulting in samples¹ $x_1, x_2, \dots$. We assume the density $p(x, \theta^*)$ of x is one of a parametric family $p(x, \theta)$ of *positive* functions, and we wish to recover θ^* from $x_1, x_2, \dots$. While $p(x, \theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over the underlying expectation E_* . Given this, the goal is to recover θ^* from the process $x_1, x_2, \dots$.

Define the *surprisal* [Samson \(1951\)](#), [Tribus \(1961\)](#)

$$S(x, \theta) = \log(1/p(x, \theta)).$$

This terminology is intuitive: low probability equals high surprisal.

Assume $S(x, \theta)$ is integrable for all θ . Then it is natural to call a minimizer θ^* of the *expected surprisal* $E_*(S(x, \theta))$ a *true parameter*, and the corresponding density $p(x, \theta^*)$ a *true density*. Equivalently, given θ^* and θ , the *relative information* is

$$I(\theta^*, \theta) = E_*(S(x, \theta)) - E_*(S(x, \theta^*)). \quad (1)$$

The relative information is the excess expected surprisal of θ over θ^* . Then θ^* is a true parameter iff $I(\theta^*, \theta) \geq 0$ for all θ .

To connect this formulation to the standard treatment of MLE consistency, we show that under an additional natural assumption, the true parameter and relative information reduce to their familiar forms.

We say θ^* is a *reference parameter* if

$$E_* \left(\frac{p(x, \theta)}{p(x, \theta^*)} \right) = 1 \quad \text{for all } \theta.$$

Then the corresponding density $p(x, \theta^*)$ is a *reference density*.

Let θ^* be a reference parameter, and define E and E_θ by

$$E_*(f(x)) = E(p(x, \theta^*)f(x)) \quad \text{and} \quad E_\theta(f(x)) = E(p(x, \theta)f(x)).$$

Then $E_\theta(1) = 1$ for all θ and the relative information takes the familiar form [Kullback & Leibler \(1951\)](#)

$$I(\theta^*, \theta) = E \left(p(x, \theta^*) \log \left(\frac{p(x, \theta^*)}{p(x, \theta)} \right) \right). \quad (2)$$

Since $j(p) = \log(1/p)$ is convex, by Jensen's inequality,

$$I(\theta^*, \theta) = E_* \left(j \left(\frac{p(x, \theta)}{p(x, \theta^*)} \right) \right) \geq j \left(E_* \left(\frac{p(x, \theta)}{p(x, \theta^*)} \right) \right) = j(E_\theta(1)) = j(1) = 0, \quad \text{for all } \theta.$$

¹We use the same letter (e.g. x) to denote both a random variable and a sample of it.

It follows that a reference density is a true density, and a reference parameter is a true parameter.

If θ is any other true parameter, then $I(\theta^*, \theta) = 0$, hence, by strict convexity of $j(p)$, there is a scalar t such that $p(x, \theta) = t \cdot p(x, \theta^*)$ with probability one.² Since θ^* is a reference parameter, $t = 1$, hence $p(x, \theta) = p(x, \theta^*)$ with probability one: every true density equals the reference density. It follows that there is at most one reference density; when there is a reference density, it is the unique true density.

We say parameters θ, θ' *share a density* if $p(x, \theta) = p(x, \theta')$ with probability one. If parameters do not share densities, there is at most one reference parameter; moreover, when there is a reference parameter, it is the unique true parameter.

In the form (2), the relative information $I(\theta^*, \theta)$ is more commonly known as Kullback-Leibler divergence [van der Vaart \(1998\)](#) or relative entropy [Cover & Thomas \(2006\)](#), [Dembo & Zeitouni \(1998\)](#) or both [Bishop \(2006\)](#), [Murphy \(2022\)](#). It is natural to regard $I(\theta^*, \theta)$ as information, following some authors [Kullback \(1959\)](#), [Schervish \(1995\)](#), since information and entropy are conceptual opposites: the [Shannon \(1948\)](#) entropy is concave, while the relative information is convex.

Returning to the general setting, the *likelihood* and (empirical) *mean surprisal* are

$$L_n(\theta) = \prod_{k=1}^n p(x_k, \theta), \quad S_n(\theta) = \frac{1}{n} \log(1/L_n(\theta)) = \frac{1}{n} \sum_{k=1}^n S(x_k, \theta).$$

The mean surprisal is also called the negative log-likelihood.

By definition, θ^* minimizes the expected surprisal. Because of this, it is natural to estimate θ^* by a minimizer of $S_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the samples, maximizing $L_n(\theta)$ over all θ . In what follows, only E_* , $p(x, \theta)$, and true parameters enter; E , E_θ , and reference parameters play no role.

3 Consistency

In addition to integrability of $S(x, \theta)$, we make four assumptions.

A (Identifiability). *There is at most one true parameter.*

A follows if the expected surprisal is strictly convex. Also, as we saw above, **A** follows if there is a reference parameter and parameters do not share densities.

A sequence of parameters *subconverges* to θ if a subsequence converges to θ . If a sequence subconverges to θ , θ is a *limit point of the sequence*. A sequence has exactly one limit point θ iff the sequence converges to θ . A subset K of the parameter space is *compact* if every sequence in K has at least one limit point in K .

A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \leq c$ are compact subsets of the parameter space, for every level c . Properness prevents minimizing sequences from escaping to infinity. Together with continuity, this forces the existence of a minimizer.

B (Properness). *There is an $N \geq 1$, depending on the samples, and a proper $F(\theta)$, not depending on the samples, such that, with probability one,*

$$F(\theta) \leq S_n(\theta), \quad \text{for all } \theta \text{ and } n \geq N. \quad (3)$$

When the parameter space is compact, the proof below shows assumption **B** is superfluous.

C (Convergence). *For every integrable $g(x)$*

$$\frac{1}{n} \sum_{k=1}^n g(x_k) \rightarrow E_*(g(x)), \quad \text{as } n \rightarrow \infty, \quad \text{with probability one.} \quad (4)$$

This implies (4) with $g(x) = S(x, \theta)$, for every θ . We will need this convergence simultaneously for all θ , in the sense

$$S_n(\theta) \rightarrow E_*(S(x, \theta)), \quad \text{as } n \rightarrow \infty, \quad \text{for all } \theta, \text{ with probability one.} \quad (5)$$

We emphasize this is not uniform convergence, since the rate of convergence may vary with parameters θ .

²With probability one is always relative to E_* .

D (Separability and Continuity). *There is a countable subset of parameters θ' of the parameter space, a function $\delta(\theta, \theta')$ satisfying $\inf_{\theta'} \delta(\theta, \theta') = 0$ for all θ , and an integrable nonnegative $g(x)$ satisfying*

$$\left| S(x, \theta) - S(x, \hat{\theta}) \right| \leq \delta(\theta, \hat{\theta})g(x), \quad \text{for all } x \text{ and } \theta, \hat{\theta}. \quad (6)$$

Under **C**, **D**, a standard separability argument establishes (5). Under **D**, the expected surprisal $E_*(S(x, \theta))$ is continuous in θ . If moreover **B**, **C** hold, then by sending $n \rightarrow \infty$ in (3), $F(\theta) \leq E_*(S(x, \theta))$ for all θ , hence the expected surprisal is proper. Hence there is a unique true parameter under **A**, **B**, **C**, and **D**. Similarly, under **B**, **D**, the mean surprisal is continuous in θ and proper, hence an MLE $\hat{\theta}_n$ exists for $n \geq N$, with probability one.

These assumptions are particularly tractable for *exponential families* Cover & Thomas (2006), Murphy (2022), Schervish (1995), van der Vaart (1998), in which $S(x, \theta)$ is essentially a sum of products of the form $f(\theta)g(x)$. In §4, we verify these assumptions for a Gaussian Markov process.

In (5), there is no presumption of uniform convergence. This is a key feature of Doob's proof. The argument instead combines properness, convergence (5), and the continuity estimate in (6) to control limit points of the MLE sequence.

Theorem (Consistency). *Let $x_1, x_2, \dots$ be a process, and let $p(x, \theta)$ be a parametric family of positive functions, such that the surprisal is integrable for all θ . Under assumptions **A**, **B**, **C**, **D**,*

1. *there is a unique true parameter θ^* ,*
2. *with probability one, an MLE sequence $\hat{\theta}_n$ exists for sufficiently large n ,*
3. *with probability one, any MLE sequence converges to the true parameter, $\hat{\theta}_n \rightarrow \theta^*$.*

Proof. The first two items were derived above. For the third item, with $g(x)$ as in (6), fix samples $x_1, x_2, \dots$ for which (3), (4), and (5) hold. We show $\hat{\theta}_n \rightarrow \theta^*$ for these samples. By (3),

$$F(\hat{\theta}_n) \leq S_n(\hat{\theta}_n) \leq S_n(\theta^*), \quad \text{for } n \geq N.$$

By (5), the right side of this last display is bounded as $n \rightarrow \infty$. Since $F(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space, and thus has limit points.

Suppose θ is a limit point of $\hat{\theta}_n$. Then $\hat{\theta}_n$ subconverges to θ . By (6),

$$S_n(\theta) - S_n(\theta^*) \leq S_n(\theta) - S_n(\hat{\theta}_n) \leq \delta(\theta, \hat{\theta}_n) \cdot \frac{1}{n} \sum_{k=1}^n g(x_k). \quad (7)$$

Since the left side of (4) is bounded as $n \rightarrow \infty$, and $\delta(\theta, \hat{\theta}_n)$ subconverges to zero, the right side of (7) subconverges to zero. By (1) and (5), the left side of (7) converges to $I(\theta^*, \theta)$, hence $I(\theta^*, \theta) \leq 0$. But $I(\theta^*, \theta) \geq 0$ since θ^* is a true parameter, hence $I(\theta^*, \theta) = 0$. Since there is only one true parameter, $\theta = \theta^*$. Thus θ^* is the only limit point, hence $\hat{\theta}_n \rightarrow \theta^*$ as $n \rightarrow \infty$. $\square$

4 Example

Let x_0 be normally distributed in $\mathbf{R}^d$ and $z_1, z_2, \dots$ an i.i.d. Gaussian sequence in $\mathbf{R}^d$ assumed independent of x_0 . Let $x_1, x_2, \dots$ be generated by the recurrence

$$x_n = A^* x_{n-1} + z_n, \quad n \geq 1.$$

Then z_n is independent of $x_1, \dots, x_{n-1}$, and $x_1, x_2, \dots$ is a Markov Gaussian process. In particular, when $A^* = 0$, $x_1, x_2, \dots$ equals the i.i.d. Gaussian sequence $z_1, z_2, \dots$.

Assume A^* is a stable matrix, μ^* is in $\mathbf{R}^d$, and Q^* is a positive definite $d \times d$ matrix. Set

$$c^* = (I - A^*)^{-1} \mu^*, \quad L^* = \sum_{n=0}^{\infty} (A^*)^n Q^* ((A^*)^t)^n, \quad (8)$$

both well-defined by stability of A^* . If the mean and variance of z_1 and x_0 are (μ^*, Q^*) and (c^*, L^*) respectively, then $x_1, x_2, \dots$ is a stationary ergodic process.

Let μ be in $\mathbf{R}^d$, Q a positive definite $d \times d$ matrix, A a $d \times d$ matrix, and let $v \cdot w$ be the dot product in $\mathbf{R}^d$. With x_0, x_1 in $\mathbf{R}^d$ and $\theta = (\mu, A, Q)$, the surprisal is

$$S(x_0, x_1, \theta) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} (x_1 - Ax_0 - \mu) \cdot Q^{-1} (x_1 - Ax_0 - \mu),$$

and the parametric family is $p(x_0, x_1, \theta) = e^{-S(x_0, x_1, \theta)}$.

Let $(x_0, x_1), (x_1, x_2), \dots$, be the *pair process*. We show $S(x_0, x_1, \theta)$ is integrable, $\theta^* = (\mu^*, A^*, Q^*)$ is the true parameter, and we verify [A](#), [B](#), [C](#), and [D](#) for the pair process, assuming stability of A^* .

For the pair process, the likelihood and mean surprisal are

$$L_n(\theta) = \prod_{k=1}^n p(x_{k-1}, x_k, \theta), \quad S_n(\theta) = \frac{1}{n} \log(1/L_n(\theta)) = \frac{1}{n} \sum_{k=1}^n S(x_{k-1}, x_k, \theta).$$

By the ergodic theorem [Billingsley \(1961\)](#), for every integrable function $g(x_0, x_1)$,

$$\frac{1}{n} \sum_{k=1}^n g(x_{k-1}, x_k) \rightarrow E_*(g(x_0, x_1)), \quad \text{as } n \rightarrow \infty, \quad \text{with probability one.} \quad (9)$$

This establishes [C](#) for the pair process.

For [D](#), with $\theta' = (\mu', A', Q')$, take $g(x_0, x_1) = 1 + |x_0|^2 + |x_1|^2$, and $\delta(\theta, \theta')$ equal to half of

$$\begin{aligned} & |\log \det Q - \log \det Q'| + |Q^{-1}\mu - (Q')^{-1}\mu'| + |Q^{-1} - (Q')^{-1}| + |\mu \cdot Q^{-1}\mu - \mu' \cdot (Q')^{-1}\mu'| \\ & + |Q^{-1}A - (Q')^{-1}A'| + |A^t Q^{-1}\mu - (A')^t (Q')^{-1}\mu'| + |A^t Q^{-1}A - (A')^t (Q')^{-1}(A')|. \end{aligned}$$

Then $\delta(\theta, \theta')$ is continuous. By standard manipulations ([6](#)) follows, establishing [D](#).

To establish [A](#) and [B](#), we summarize the necessary matrix manipulations in the following proposition. Let A and B be symmetric matrices. Throughout we say $A \geq B$ if $v \cdot Av \geq v \cdot Bv$ for all vectors v . Then the trace is monotone in the sense $A \geq B$ and $Q > 0$ imply $\text{trace}(Q^{-1}A) \geq \text{trace}(Q^{-1}B)$.

Proposition. For $Q^* > 0$ and c scalar, let $\alpha = e^{-2c-1} \det(2\pi e Q^*)$ and

$$F_*(Q, Q^*) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \text{trace}(Q^{-1}Q^*), \quad Q > 0.$$

Then $F_*(Q, Q^*)$ is minimized iff $Q = Q^*$, and $F_*(Q, Q^*) \leq c$ implies $\alpha Q^* \leq Q \leq Q^*/\alpha$.

Proof. Let $h(\lambda) = \log \lambda + 1/\lambda$, $\lambda > 0$. Then $h(\lambda)$ is minimized iff $\lambda = 1$ and $h(\lambda)$ is proper in the sense $h(\lambda) \leq \log(1/\alpha)$ implies $\alpha \leq \lambda \leq 1/\alpha$. Write $Q^* = S^2$ with $S > 0$, and define Z by $Q = SZS$. Suppose $F_*(Q, Q^*) \leq c$ and let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Z . Then the proposition follows from

$$\log(1/\alpha) - 1 \geq 2F_*(Q, Q^*) - 2F_*(Q^*, Q^*) = \log \det Z + \text{trace}(Z^{-1} - I) = \sum_{i=1}^d (h(\lambda_i) - 1). \quad \square$$

By monotonicity of the trace, $F_*(Q, Q^*)$ is analogously monotone in Q^* . Given vectors $v = (s_1, s_2, \dots, s_d)$ and $w = (t_1, t_2, \dots, t_d)$, the *tensor product* $v \otimes w$ is the matrix whose ij -th entry is $s_i t_j$. Then $z \cdot Q^{-1}z = \text{trace}(Q^{-1}(z \otimes z))$.

Let $D = A^* - A$ be the difference and let $v = (\mu^* - \mu) + Dc^*$. The mean and variance of

$$x_k - Ax_{k-1} - \mu = z_k + Dx_{k-1} - \mu = (z_k - \mu^*) + D(x_{k-1} - c^*) + v \quad (10)$$

are v and $Q^* + DL^*D^t$. Here we used $c^* - A^*c^* = \mu^*$ ([8](#)). If

$$f_0(\mu, A) = Q^* + DL^*D^t + v \otimes v, \quad F_0(\theta) = F_*(Q, f_0(\mu, A)),$$

we conclude

$$E_*(S(x_0, x_1, \theta)) = F_0(\theta) = F_0(Q, Q^*) + F_0(Q, DL^*D^t) + \frac{1}{2}v \cdot Q^{-1}v.$$

The first term is uniquely minimized at $Q = Q^*$, $DL^*D^t \geq 0$ is uniquely minimized at $A = A^*$, and $v \cdot Q^{-1}v$ is uniquely minimized at (μ^*, A^*) . This establishes A. Moreover, since the second and third terms are nonnegative, inspecting $F_0(\mu, A, Q) \leq c$ implies Q , then A , then μ are confined to compact subsets of their domains, hence $F_0(\mu, A, Q)$ is proper.

It remains to establish B. With $\hat{\mu}_n$ equal to the sample mean $(\sum_{k=1}^n z_k)/n$, define

$$\hat{c}_n = \frac{1}{n} \sum_{k=1}^n x_{k-1}, \quad \hat{L}_n = \frac{1}{n} \sum_{k=1}^n (x_{k-1} - c^*) \otimes (x_{k-1} - c^*), \quad \hat{C}_n = \frac{1}{n} \sum_{k=1}^n (z_k - \mu^*) \otimes (x_{k-1} - c^*).$$

Let

$$Q_n(\mu, A) = \frac{1}{n} \sum_{k=1}^n (x_k - Ax_{k-1} - \mu) \otimes (x_k - Ax_{k-1} - \mu).$$

Then $S_n(\theta) = F_*(Q, Q_n(\mu, A))$.

By (10), with $w = \hat{\mu}_n - \mu^*$ and $u = D(\hat{c}_n - c^*)$, the trinomial expansion of $Q_n(\mu, A)$ is the sum of three square terms and three cross terms,

$$Q_n(\mu, A) = Q_n(\mu^*, A^*) + D\hat{L}_nD^t + v \otimes v + (u \otimes v + v \otimes u) + (w \otimes v + v \otimes w) + (D\hat{C}_n^t + \hat{C}_nD^t).$$

For $0 < \epsilon < 1/3$, set

$$f_\epsilon(\mu, A) = (1 - 3\epsilon)Q^* + (1 - 3\epsilon)DL^*D^t + (1 - 2\epsilon)v \otimes v,$$

and $F_\epsilon(\theta) = F_*(Q, f_\epsilon(\mu, A))$. Then, as before, $F_\epsilon(\theta)$ is proper.

By (9), with probability one, $Q_n(\mu^*, A^*)$, $\hat{L}_n$, $\hat{c}_n$, and $\hat{\mu}_n$ converge to Q^* , L^* , c^* , and μ^* . Since $z_k = x_k - A^*x_{k-1}$, by (9), $\hat{C}_n \rightarrow E_*((z_1 - \mu^*) \otimes (x_0 - c^*))$ with probability one. By independence of z_1 and x_0 , $E_*((z_1 - \mu^*) \otimes (x_0 - c^*)) = 0$. Thus $\hat{C}_n \rightarrow 0$ with probability one. It follows there is $N \geq 1$, depending on the samples, such that, with probability one,

$$\begin{aligned} Q_n(\mu^*, A^*) &\geq (1 - \epsilon)Q^*, & \hat{L}_n &\geq (1 - \epsilon)L^*, & (\hat{c}_n - c^*) \otimes (\hat{c}_n - c^*) &\leq \epsilon^2 L^*, \\ (\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) &\leq \epsilon^2 Q^*, & \hat{C}_n(L^*)^{-1}\hat{C}_n^t &\leq \epsilon^2 Q^*, \end{aligned} \quad \text{for } n \geq N.$$

Fix samples $x_1, x_2, \dots$ for which these hold. We establish (3) for these samples and $F(\theta) = F_\epsilon(\theta)$.

Using the matrix inequality

$$\pm(VW^t + WV^t) \leq \epsilon VLV^t + \frac{1}{\epsilon}WL^{-1}W^t,$$

one can check the negative of the sum of the cross terms is bounded above by $2\epsilon f_0(\mu, A)$ for $n \geq N$, hence

$$Q_n(\mu, A) \geq f_\epsilon(\mu, A), \quad \text{for all } \mu, A \text{ and } n \geq N.$$

Since $S_n(\theta) = F_*(Q, Q_n(\mu, A))$, (3) follows. This establishes B.

The MLE $\hat{\theta}_n$ is obtained by doing least squares of $Q_n(\mu, A)$ over (μ, A) , leading to the (multivariate) Yule–Walker estimator Yule (1927), Walker (1931).

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